

ELEC4631 Quiz 1

Instructions:

- This is an assessment item. You may not discuss the quiz problems or post your solutions on the Moodle student's forum. Violating this will be considered as academic misconduct.
- Submitted solutions must be your own work, not gained through plagiarism (copying someone's else work). Plagiarism is an academic misconduct and can result in zero marks for both the plagiator and the source.
- **All analytical workings must be shown in your solution. No workings and demonstration of the solution steps, no marks. Solutions by Matlab only will receive no marks.**
- You will submit your assignment via Moodle. Only submissions in pdf format will be accepted. Your assignment may be typed up or handwritten and scanned. Handwritten and scanned solutions must be clear and readable, and scans should not be in very high resolution that results in a large file size. Unreadable submissions will not be marked.
- Due by date: Wednesday 21/06/2023, 5 pm. Penalty for late submissions apply per the course outline.

Total marks for this quiz: 100

1. **(20 marks total)** Let the matrix A be given by

$$A = \begin{bmatrix} 5 & 1 & 0 \\ -1 & 5 + 2\delta & \delta + 1 \\ 0 & 1 - \delta & 5 \end{bmatrix},$$

for some unknown constant δ with $0 < |\delta| < 1$ (in particular, $\delta \neq 0$). Answer the following questions:

- (8 marks) Determine the characteristic polynomial and all eigenvalues of A . The solution should depend on δ .
- (10 marks) Determine the geometric multiplicity of all the eigenvalues of A
- (2 marks) Determine if the matrix is diagonalisable or not.

2. (25 marks total) Consider the matrix

$$A = \begin{bmatrix} 2 & \delta & 0 \\ -5 & -2 & -1 \\ 0 & 4 & 2 \end{bmatrix},$$

for some unknown constant $\delta > 0$. Find an invertible matrix V such that $V^{-1}AV = D$ for a diagonal matrix D and determine D . The solution should depend on δ .

3. (40 marks) Consider the electrical circuit shown in Fig. 1 that has a resistor with resistance $R > 0$, an inductor with inductance $L > 0$, a capacitor with capacitance $C > 0$, a nonlinear element Z that has a nonlinear voltage-current relationship given by:

$$i_4 = g(v_4) = \begin{cases} e^{2v_4} - 1, & v_4 \geq 0 \\ \frac{1}{2}(1 - e^{-4v_4}), & v_4 < 0 \end{cases},$$

and a voltage source with voltage u as its control signal (input).

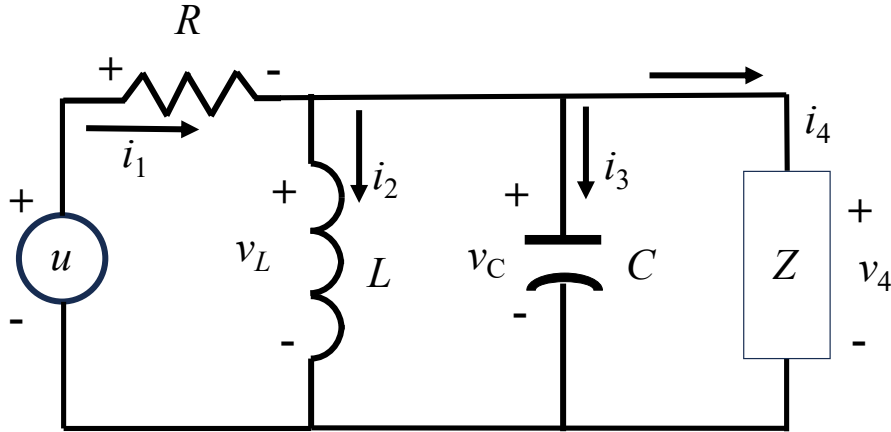


Figure 1: Electrical circuit for Q3.

Let the state-space of the circuit be $\mathbb{R}^2$ with states $x = (v_C, i_2)^T$, where the v_C is the voltage across the capacitor and i_2 is the current flowing across the inductor. Derive the controlled ODE

$$\dot{x} = f(x, u),$$

that describes the dynamics of the circuit.

4. (15 marks) Consider again the circuit in Problem 3. If the input u is small, $|u(t)| \ll 1$ for all $t \geq 0$, and the initial state x_0 is also small, then $\|x(t)\|$ is small for all $t \geq 0$. In this case the dynamics of the circuit becomes approximately linear. Derive an LTI system

$$\dot{x} = Ax + Bu,$$

for appropriate constant matrices $A \in \mathbb{R}^{2 \times 2}$ and $B \in \mathbb{R}^{2 \times 1}$, that approximately describes the dynamics of the circuit. **Hint:** Use the approximation $g(v_C) \approx 2v_C$